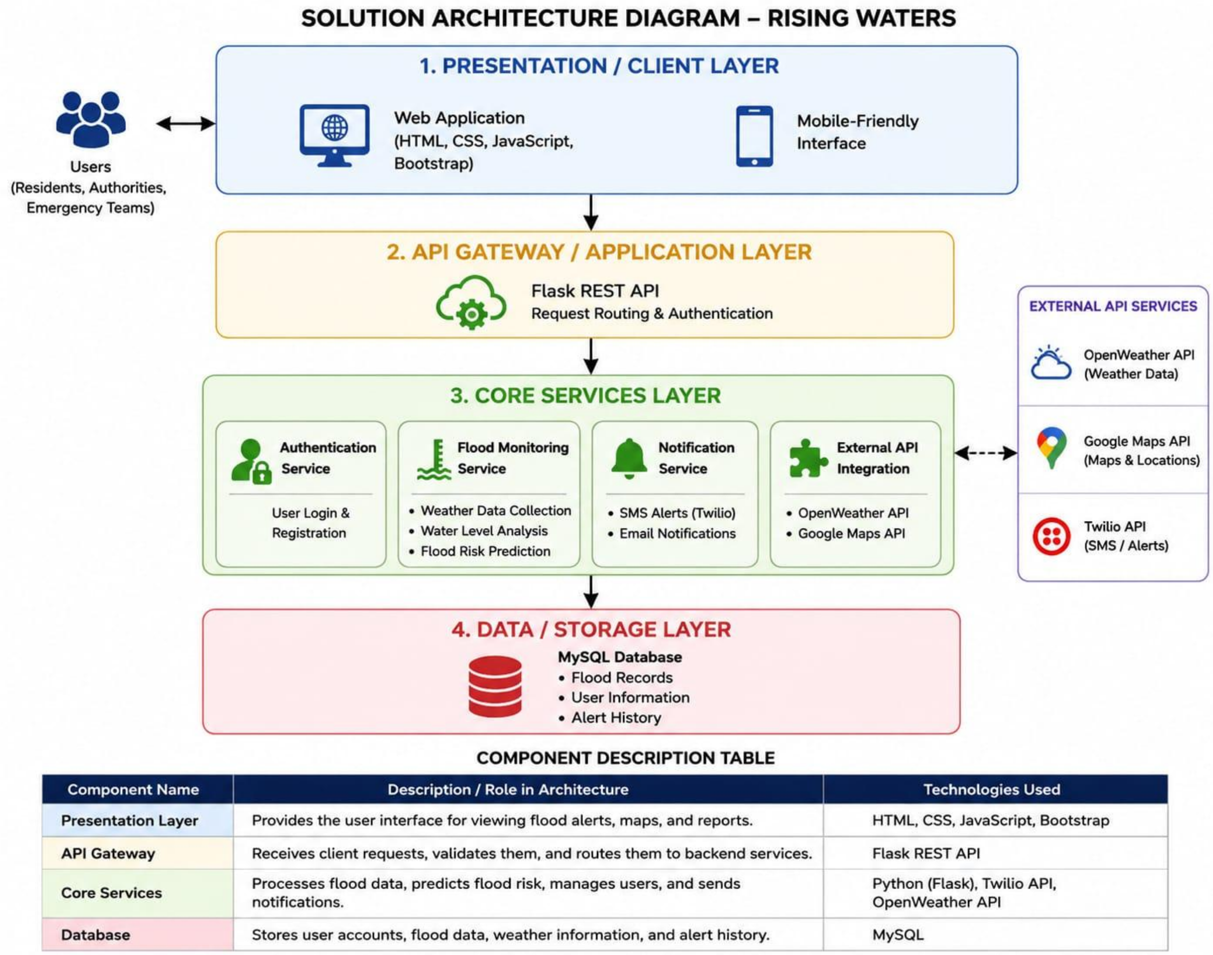
Solution Architecture

Date	29 June 2026
Team ID	xxxxxx
Project Name	RISING WATERS
Maximum Marks	5 Marks

Solution Architecture Diagram:

The Rising Waters solution architecture follows a layered architecture consisting of the Presentation Layer, API Gateway, Application Services, External API Integration, and Data Storage Layer. Users access the system through a responsive web interface. Requests are routed through the Flask REST API to core services that handle user authentication, flood monitoring, weather analysis, and emergency notifications. The application integrates with OpenWeather API, Google Maps API, and Twilio for real-time weather data, flood mapping, and SMS alerts. All user information, flood records, and alert history are securely stored in a MySQL database, ensuring reliable, scalable, and secure system performance.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface for viewing flood alerts, maps, and reports.	*HTML, CSS, JavaScript, Bootstrap*
API Gateway	Receives client requests, validates them, and routes them to backend services.	Flask REST API
Core Services	*Processes flood data, predicts flood risk, manages users, and sends notifications.*	Python (Flask), Twilio API, OpenWeather API
Database	stores user accounts, flood data, weather information, and alert history.	MySQL